

NeRF: Hierarchical Volume Sampling

Continuous

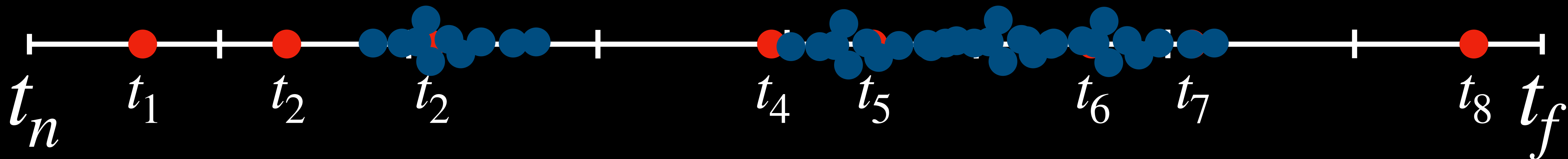
$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{r}(t)) \mathbf{c}(\mathbf{r}(t), \mathbf{d}) dt$$

$$T(t) = \exp \left(- \int_{t_n}^t \sigma(\mathbf{r}(s)) ds \right)$$

Discrete

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^N T_i (1 - \exp(-\sigma_i \delta_i)) \mathbf{c}_i$$

$$T(t) = \exp \left(- \sum_{j=1}^{i-1} \sigma_j \delta_j \right)$$



NeRF: Optimization

$$\mathcal{L} = \sum_{\mathbf{r} \in \mathcal{R}} \left[\underbrace{\left\| \hat{C}_c(\mathbf{r}) - C(\mathbf{r}) \right\|_2^2}_{\text{Coarse}} + \underbrace{\left\| \hat{C}_f(\mathbf{r}) - C(\mathbf{r}) \right\|_2^2}_{\text{Fine}} \right]$$

